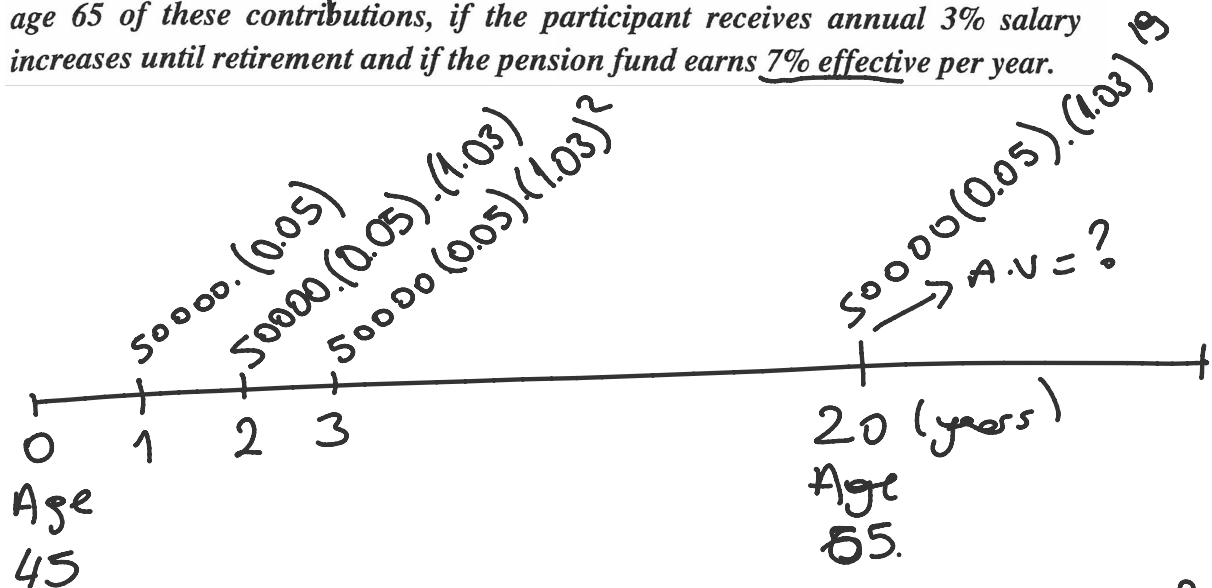


Example 4.14 A participant in a pension plan aged 45 currently earns \$50,000 per year. This participant plans to contribute 5% of salary at the end of each year into the pension fund for 20 years. Find the accumulated value at age 65 of these contributions, if the participant receives annual 3% salary increases until retirement and if the pension fund earns 7% effective per year.



$$\begin{aligned}
 A.V. &= 50000 \cdot (0.05) \left[1 \cdot (1.07)^{19} + \frac{(1.03)^1 \cdot (1.07)^{18}}{(1.07)^{19}} + (1.03)^2 \cdot (1.07)^{17} + \dots \right. \\
 &\quad \left. \dots + (1.03)^{19} \right] \\
 &= 2500 \cdot (1.07)^{19} \left[1 + \frac{1.03}{1.07} + \left(\frac{1.03}{1.07} \right)^2 + \dots + \left(\frac{1.03}{1.07} \right)^{19} \right] \\
 &\quad \text{a geometric series with the common factor } \frac{1.03}{1.07} \\
 &\quad \cdot \frac{1-r^n}{1-r}, \quad r = \text{common ratio}
 \end{aligned}$$

$$= 2500 \cdot (1.07)^{19} \cdot \frac{1 - \left(\frac{1.03}{1.07} \right)^{20}}{1 - \frac{1.03}{1.07}} = 128973 //$$

The accumulated value of the contributions made by this participant at the end of 20 years (i.e. at age 65) is

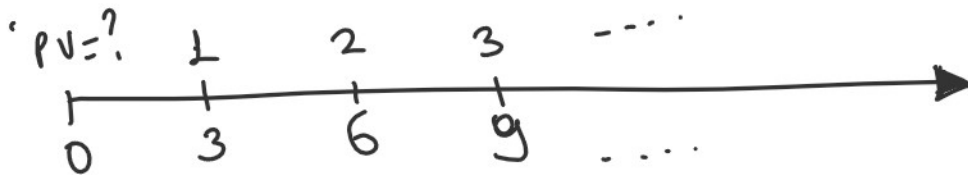
$$50,000(.05) \left[(1.07)^{19} + (1.03)(1.07)^{18} + \dots + (1.03)^{18}(1.07) + (1.03)^{19} \right]$$

which can be evaluated using formula (4.34) to give

$$2500 \frac{(1.07)^{20} - (1.03)^{20}}{.07 - .03} = \$128,973$$

to the nearest dollar.

Example 4.15 Find the present value of a perpetuity which pays 1 at the end of the third year, 2 at the end of the sixth year, 3 at the end of the ninth year, ...



$$v = \frac{1}{1+i}$$

$$\frac{1-r^n}{1-r} //$$

$$\checkmark P.V = 1 \cdot v^3 + 2 \cdot v^6 + 3 \cdot v^9 + \dots$$

$$\times = v^3 (1 + 2 \cdot v^3 + 3 \cdot v^6 + \dots)$$

this is not a geometric series.

$$\checkmark + v^3 \cdot (P.V) = v^6 + 2 \cdot v^9 + 3 \cdot v^{12} + \dots \quad \times (-1)$$

$$(P.V) [1 - v^3] = v^3 + v^6 + v^9 + \dots$$

$$\frac{1-r^n}{1-r} = \frac{1}{1-r}$$

$$(P.V) (1 - v^3) = v^3 [1 + v^3 + v^6 + \dots]$$

This is a geo. series with $r = v^3$.

$$P.V = \frac{v^3}{1-v^3} \cdot \frac{1}{1-v^3} = \frac{v^3}{(1-v^3)^2} //$$

Denote the present value of this perpetuity by A . Then

$$A = v^3 + 2v^6 + 3v^9 + \dots$$

and

$$v^3 A = v^6 + 2v^9 + \dots$$

Now subtracting the second equation from the first

$$A(1-v^3) = v^3 + v^6 + v^9 + \dots = \frac{v^3}{1-v^3}$$

so that

$$A = \frac{v^3}{(1-v^3)^2}$$

Example 4.16 Find the accumulated value at the end of ten years of an annuity in which payments are made at the beginning of each half-year for five years. The first payment is \$2000, and each payment is 98% of the prior payment. Interest is credited at 10% convertible quarterly.

Timeline diagram showing payments at the beginning of each half-year (0, 0.5, 1, 1.5, 2, ..., 4.5, 5 years) for 10 years. The first payment is 2000, and subsequent payments are 2000(0.98), 2000(0.98)², 2000(0.98)³, ..., 2000(0.98)⁹. The interest rate is $i^{(4)} = 0.10$.

Effective interest rate per quarter: $j = \frac{i^{(4)}}{4} = 0.025$

Accumulated Value (A.V.) calculation:

$$A.V. = 2000 \left[(1.025)^{40} + (0.98) \cdot (1.025)^{38} + (0.98)^2 \cdot (1.025)^{36} + (0.98)^3 \cdot (1.025)^{34} + \dots + (0.98)^9 \cdot (1.025)^{22} \right]$$

This is a geo. series with $r = \frac{0.98}{(1.025)^2}$

$$A.V. = 2000 \cdot (1.025)^{40} \left[1 + \frac{0.98}{(1.025)^2} + \left(\frac{0.98}{(1.025)^2} \right)^2 + \dots + \left(\frac{0.98}{(1.025)^2} \right)^9 \right]$$

Final result: $40 \quad 1 - r^{10} \quad 0.98$

$$= 2000 \cdot (1.025)^{40} \cdot \frac{1 - r^{10}}{1 - r} \quad \text{where} \quad r = \frac{0.98}{(1.025)^2} \quad (1.025)$$

We count periods in quarters of a year. The accumulated value is

$$\begin{aligned} & 2000 \left[(1.025)^{40} + (.98)(1.025)^{38} + (.98)^2(1.025)^{36} + \dots + (.98)^9(1.025)^{22} \right] \\ &= 2000 \frac{(1.025)^{40} - (.98)^{10}(1.025)^{20}}{1 - (.98)(1.025)^{-2}} \\ &= \$40,052 \end{aligned}$$

to the nearest dollar.

3 If $a_{\overline{n}|} = x$ and $a_{\overline{2n}|} = y$, express d as a function of x and y .

We are given $a_{\overline{n}|} = \frac{1 - v^n}{i} = x$, so that $v^n = 1 - ix$.

$$a_{\overline{2n}|} = \frac{1 - v^{2n}}{i} = y, \quad \text{so that } v^{2n} = 1 - iy.$$

But $v^{2n} = (v^n)^2$ so that $1 - iy = (1 - ix)^2$.

$$x^2 i^2 - (2x - y)i = 0 \quad \longrightarrow \quad i = \frac{2x - y}{x^2}.$$

$$d = \frac{i}{1 + i} = \frac{2x - y}{x^2 + 2x - y}.$$

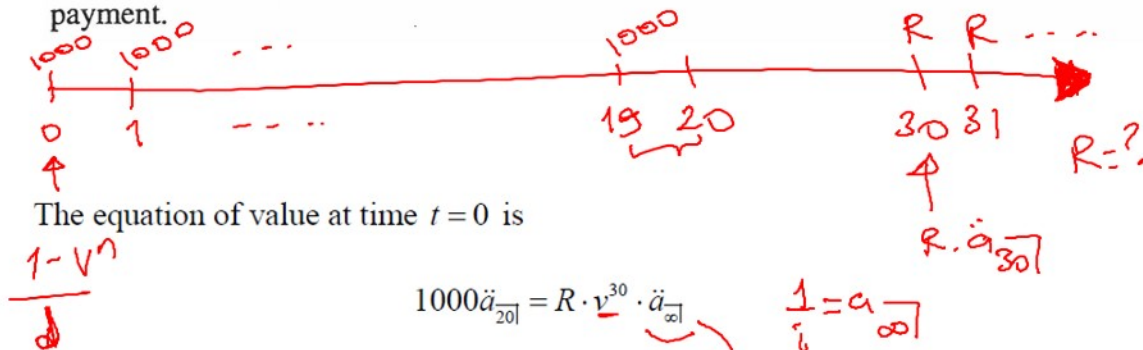
4 Find the present value of payments of \$200 every six months starting immediately and continuing through four years from the present, and \$100 every six months thereafter through ten years from the present, if $i^{(2)} = .06$.

The semiannual interest rate is $j = .06 / 2 = .03$.

The present value of the payments is

$$100(\ddot{a}_{\overline{20}|} + \ddot{a}_{\overline{9}|}) = 100(15.87747 + 8.01969) = \$2389.72.$$

- 6 Deposits of \$1000 are placed into a fund at the beginning of each year for the next 20 years. After 30 years annual payments commence and continue forever, with the first payment at the end of the 30th year. Find an expression for the amount of each payment.



$$1000\ddot{a}_{\overline{20}|} = R \cdot v^{30} \cdot \ddot{a}_{\overline{\infty}|}$$

$$1000 \frac{1-v^{20}}{d} = R \cdot v^{30} \frac{1}{d}$$

so that

$$\begin{aligned} R &= 1000 \frac{1-v^{20}}{v^{30}} = 1000(1-v^{20})(1+i)^{30} \\ &= 1000[(1+i)^{30} - (1+i)^{10}]. \end{aligned}$$

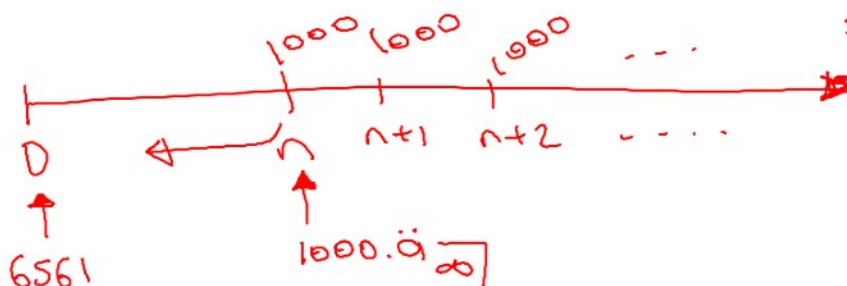
- 7 A deferred perpetuity-due begins payments at time n with annual payments of \$1000 per year. If the present value of this perpetuity-due is equal to \$6561 and the effective rate of interest $i = 1/9$, find n .

We are given $i = \frac{1}{9}$ so that $d = \frac{i}{1+i} = \frac{1}{10}$.

The equation of value at time $t = 0$ is

$$6561 = 1000v^n \ddot{a}_{\infty} \quad \text{or} \quad 6.561 = \frac{(1-d)^n}{d} = \frac{(1-.1)^n}{.1}$$

Therefore, $(.9)^n = (.1)(6.561) = .6561$ and $n = 4$.



$$\begin{aligned} v &= 1-d \\ (.9)^n &= 0.6561 \\ \ln(.9)^n &= \ln(0.6561) \\ n &= \frac{\ln(0.6561)}{\ln(0.9)} \\ &= 4 // \end{aligned}$$

8

- 8 A loan of \$1000 is to be repaid by annual payments of \$100 to commence at the end of the fifth year and to continue thereafter for as long as necessary. Find the time and amount of the final payment, if the final payment is to be larger than the regular payments. Assume $i = 4 \frac{1}{2}\%$.

At time $t = 0$ we have the equation of value

$$1000 = 100(a_{\overline{n}|} - a_{\overline{4}|}) \quad \text{or}$$

$$a_{\overline{n}|} = 10 + a_{\overline{4}|} = 13.5875 \quad \text{at } i = .045.$$

$$n = 21.4759$$

We now use time $t = 21$ as the comparison date to obtain

$$1000(1.045)^{21} = 100s_{\overline{17}|} + \text{Final Pay}$$

$$\begin{aligned} \text{Final Pay} &= 1000(1.045)^{21} - 100s_{\overline{17}|} \\ &= 2520.2412 - 100(24.74171) = 46.07 \end{aligned}$$

Thus, the balloon payment is

$$100 + 46.07 = \$146.07 \quad \text{at time } t = 21.$$

- 9 A beneficiary receives a \$10,000 life insurance benefit. If the beneficiary uses the proceeds to buy a 10-year annuity-immediate, the annual payout will be \$1538. If a 20-year annuity-immediate is purchased, the annual payout will be \$1072. Both calculations are based on an annual effective interest rate of i . Find i .

We have the following equation of value

$$10,000 = 1538a_{\overline{10}|} = 1072a_{\overline{20}|}.$$

$$\text{Thus } 1538(1-v^{10}) = 1072(1-v^{20}) = 1072(1-v^{10})(1+v^{10}),$$

$$\text{so that } 1+v^{10} = \frac{1538}{1072} \text{ or } v^{10} = .43470.$$

Solving for i , we obtain

$$1+i^{10} = .43470 \text{ and } i = (.43470)^{-1/10} - 1 = .0869, \text{ or } 8.69\%.$$

- 10 Find the accumulated value 18 years after the first payment is made of an annuity on which there are 8 payments of \$2000 each made at two-year intervals. The nominal rate of interest convertible semiannually is 7%. Answer to the nearest dollar.

$$(1+j) = \left(1 + \frac{i^{(2)}}{2}\right)^4$$

The nominal rate of interest convertible once every two years is j , so that

$$j = \frac{i^{(2)}}{2} \quad 1+j = \left(1 + \frac{.07}{2}\right)^4 \text{ and } j = (1.035)^4 - 1 = .14752.$$

The accumulated value is taken 4 years after the last payment is made, so that

Assume it is an ann-im.

$$2000s_{\overline{8}|j}(1+j)^2 = 2000(13.60268)(1.31680) = \$35,824 \text{ to the nearest dollar.}$$

